



ENGINEERING MATHEMATICS-II

Department of Science and Humanities



ENGINEERING MATHEMATICS-II

UNIT 1: INTEGRAL CALCULUS

Class 2: applications of double integral

GEOMETRICAL APPLICATIONS OF DOUBLE INTEGRALS

1. **Volume V** of the solid S
2. The **area** of the region R is given by
3. The **Average value** of a function of two variables over a region R is
4. If f and its first partial derivatives are continuous on the closed region R in the xy -plane, then the **area of the surface S** given by $z = f(x, y)$ over R is defined as

Surface area =

PHYSICAL APPLICATIONS OF DOUBLE INTEGRALS

1. The mass of the Lamina

If $\delta(x, y)$ describes the density of a lamina defined by a planar region D , then the mass of D is given by the double Integral

2. The center of mass of Lamina

The coordinates of the center of mass of a Lamina D with the density $\delta(x, y)$ are given by

$$\bar{x} = \frac{\iint_D x \delta(x, y) dA}{\iint_D \delta(x, y) dA} \quad \bar{y} = \frac{\iint_D y \delta(x, y) dA}{\iint_D \delta(x, y) dA}$$

3. Given a joint probability density function f is a function of two independent variables x and y defined on a domain D , if R is some subregion of D , then the probability that is in R is given by

PHYSICAL APPLICATIONS OF DOUBLE INTEGRALS

The moment of inertia of a lamina about the x -axis is defined by the formula

$$I_x = \iint_R y^2 \rho(x, y) dA.$$

Similarly, the moment of inertia of a lamina about the y -axis is given by

$$I_y = \iint_R x^2 \rho(x, y) dA.$$

The polar moment of inertia is

$$I_0 = \iint_R (x^2 + y^2) \rho(x, y) dA.$$

PHYSICAL APPLICATIONS OF DOUBLE INTEGRALS

Charge of a Plate

Suppose electrical charge is distributed over a region which has area R in the xy -plane and its charge density is defined by the function $\sigma(x, y)$. Then the total charge Q of the plate is defined by the expression

$$Q = \iint_R \sigma(x, y) dA.$$

AREA ENCLOSED BY THE PLANE CURVE

Cartesian form

Let AB and DC be two curves $y = f_1(x)$ and $y = f_2(x)$ respectively. Also Let AD and BC be the ordinates $x = x_1$ and $x = x_2$ respectively. Then the area enclosed by the curves $y = f_1(x)$ and $y = f_2(x)$ And ordinates $x=x_1$ and $x = x_2$ is ABCD.

$$\text{Area ABCD} = \int_{x_1}^{x_2} \int_{y=f_1(x)}^{y=f_2(x)} dy dx$$

AREA ENCLOSED BY THE PLANE CURVE

Find the area enclosed by the curve $y^2=4ax$ and the line $x+y=3a$, $y=0$.

Given $x+y=3a$

(1)

$y^2=4ax$

(2)

We have $x+y=3a$

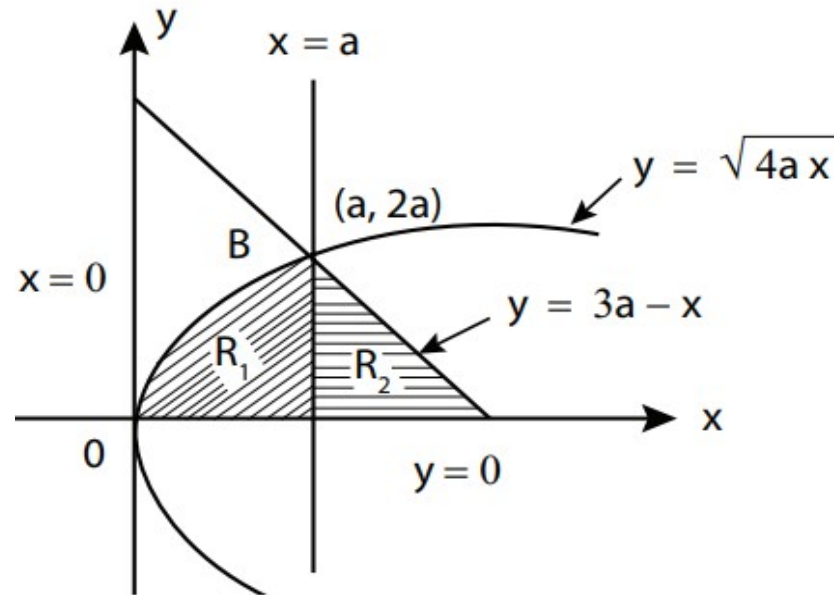
$\Rightarrow y=3a-x$

Substitute $y=3a-x$ in (2)

We get $x=a$, $x=9a$

Substitute $x=a$ in (1)

We get $y=2a$



Area split into two regions namely R_1 and R_2 .

AREA ENCLOSED BY THE PLANE CURVE

$$\begin{aligned}
 \text{Area of R1} &= \int_0^a \int_0^{\sqrt{4ax}} dy \, dx \\
 &= \int_0^a (y)_0^{\sqrt{4ax}} dx \\
 &= \int_0^a (\sqrt{4ax}) \, dx \\
 &= 2\sqrt{a} \int_0^a x^{1/2} dx \\
 &= 2\sqrt{a} \left(\frac{x^{3/2}}{3/2} \right)_0^a \\
 &= \frac{4a^2}{3}
 \end{aligned}$$

area of R1 = 9 sq. units

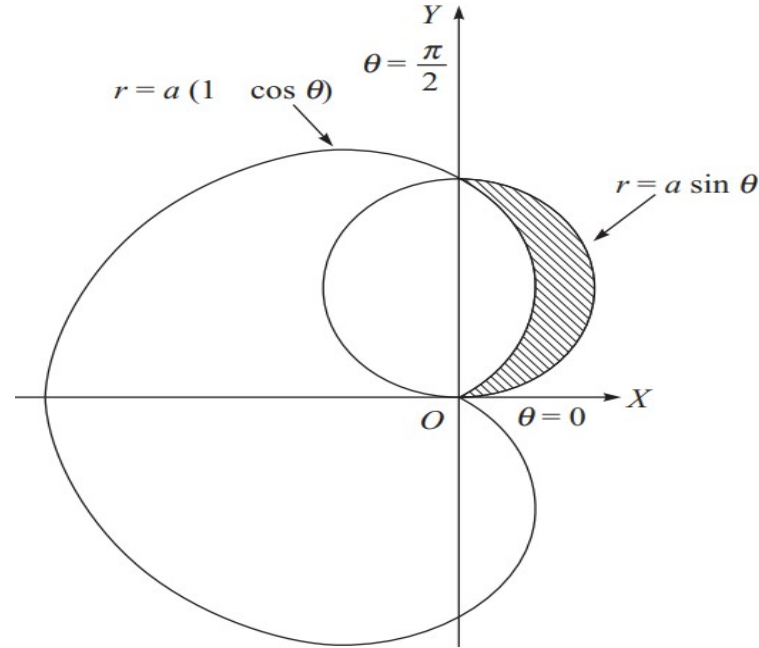
$$\begin{aligned}
 \text{Area of R2} &= \int_a^{3a} \int_0^{3a-x} dy \, dx \\
 &= \int_a^{3a} (y)_0^{3a-x} dx \\
 &= \int_a^{3a} (3a-x) dx \\
 &= \left(\frac{(3a-x)^2}{-2} \right)_a^{3a} \\
 &= 2a^2
 \end{aligned}$$

Area = Area of R1 + Area of R2

$$\begin{aligned}
 &= \frac{4a^2}{3} + 2a^2 \\
 &= \frac{10a^2}{3}
 \end{aligned}$$

AREA ENCLOSED BY THE PLANE CURVE (POLAR FORM)

Find the area lying inside the circle
 $r = a \sin \theta$ and outside the cardioid
 $r = a (1 - \cos \theta)$ using double Integration.



AREA ENCLOSED BY THE PLANE CURVE (POLAR FORM)

$$\begin{aligned}\text{Required area} &= \int_0^{\pi/2} \int_{a(1-\cos\theta)}^{a\sin\theta} r \, dr \, d\theta \\ &= \int_0^{\pi/2} \left. \frac{r^2}{2} \right|_{a(1-\cos\theta)}^{a\sin\theta} d\theta \\ &= \frac{1}{2} \int_0^{\pi/2} a^2 \left[\sin^2 \theta - (1 - \cos \theta)^2 \right] d\theta \\ &= \frac{a^2}{2} \int_0^{\pi/2} (\sin^2 \theta - 1 - \cos^2 \theta + 2 \cos \theta) d\theta \\ &= \frac{a^2}{2} \int_0^{\pi/2} (-2 \cos^2 \theta + 2 \cos \theta) d\theta = a^2 \left[-\frac{1}{2} \cdot \frac{\pi}{2} + 1 \right] = a^2 \left(1 - \frac{\pi}{4} \right)\end{aligned}$$

VOLUME OF THE SOLID

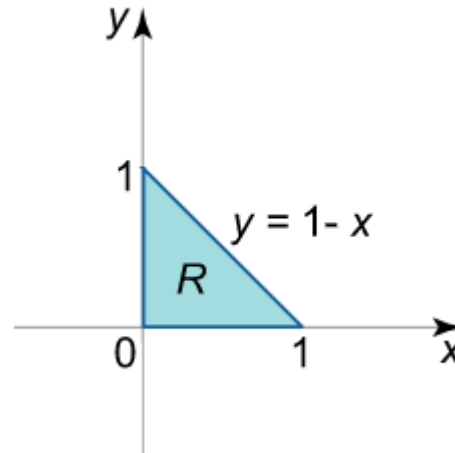
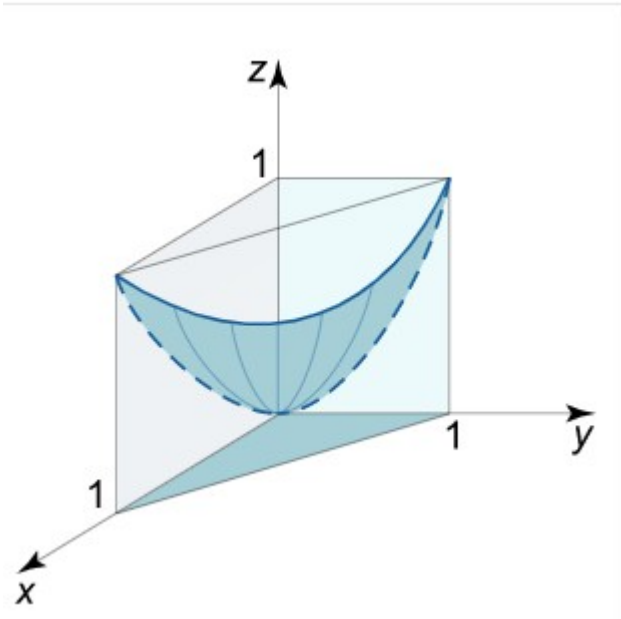
A double integral can be used to find the volume of a solid region that lies between the xy -plane and the surface given by $z = f(x, y)$.

If f is integrable over a plane region R and $f(x, y) \geq 0$ for all (x, y) in R , then the volume of the solid region that lies above R and below the graph of f is defined as

$$V = \iint_R f(x, y) \, dA.$$

VOLUME OF THE SOLID

Describe the solid whose volume is given by the integral





THANK YOU

